Eric Maniraguha

Data Engineer · Database & Analytics Engineer

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About Me

Data Engineer with **10+ years of experience** building databases, ETL/ELT pipelines, BI dashboards, and data quality systems across **fintech, healthcare, humanitarian, research, and public-sector** organizations. **MSc in Information Technology (Data Science & Machine Learning), Carnegie Mellon University Africa.**

Core stack: SQL · PostgreSQL · Oracle PL/SQL · MySQL · Python · PySpark · Apache Airflow · dbt · Kafka · Power BI · Grafana · Docker · AWS

> ⚠️ **Confidentiality note:** Most of my work was delivered for client and partner organizations. I am **not permitted to share their repositories, source code, or data**. To give a clear and honest picture of that work, this repository presents **historical project information** instead: what each organization needed, what I built, and the results.

Key Results

Result	Where
98% data consistency across integrated datasets	Dussoft Engineering
23% gain in operational efficiency through pipeline automation	Dussoft Engineering
13% reduction in data processing time	Dussoft Engineering
Database systems supporting **90+** operational units	BrightSeed Hub
18,200-record golden dataset; **30** potential ghost employees flagged	HR Data Integration Pipeline

Project History

Period	Organization · Role	What I Delivered	Technologies
2025 – Present	**BrightSeed Hub** Database Engineering & Systems Analyst Consultant	Database design and optimization for financial applications; migration, reconciliation, and automated data quality checks; Grafana monitoring; RBAC, backup, and recovery	PostgreSQL, SQL, Python, Grafana

Period	Organization · Role	What I Delivered	Technologies
2026	**eFiche** Digital Health Implementation Consultant	Led EHR data migration: validation, transformation, and integrity checks; ETL and interoperability support; English–French localization	SQL, ETL, REST APIs
2026	**DRA VANUA Group** ERP Project Lead	End-to-end ERP data platform covering sales, inventory, HR, payroll, finance, and assets; RBAC, audit logging, KPI dashboards, Dockerized deployment	MySQL, Node.js, Sequelize, Grafana, Docker
2023 – 2025	**Upanzi Network: Mojaloop** Research Associate, CyLab Africa	Multi-currency ledger data model, reporting dashboards, Grafana and Loki observability, query optimization	PostgreSQL, MySQL, Grafana, Loki, Prometheus
2023 – 2025	**CyLab Africa** Data Scientist & ICT Systems Analyst	Security telemetry ingestion pipeline, event categorization for threat analytics, data platform governance	Python, PySpark, PostgreSQL, Elasticsearch
2023 – 2025	**Dussoft Engineering** Technical Focal Person, Data & Databases	Database administration, Airflow ETL with reconciliation, migration and recovery, runbooks, Power BI and Kibana dashboards	Oracle, PL/SQL, PostgreSQL, Airflow, Power BI
2024 – 2025	**Malaria Surveillance Analytics** Kamabuye Health Center	Geospatial pipelines and GIS dashboards for hotspot analysis	MongoDB, PostgreSQL, Airflow, Kibana
2023	**C4IR Rwanda** AI Developer	LLM document pipeline: chunking, embeddings, vector search, RAG	Python, LLMs, vector databases
2019 – 2022	**Dussoft: Humanitarian Market Intelligence** Data Engineer & Team Lead	Multi-source ETL/ELT, deduplication, data quality checks, food-price and supply-chain Power BI dashboards	PostgreSQL, Airflow, AWS S3, Redshift, Power BI
2015 – 2019	**Tech-Connect Business Park** ICT Systems & E-Business Consultant	Database design and administration, e-payment workflows, RBAC data access, system integration	SQL, relational databases
ADD YEARS	**FinTech Core & Payment Engine** Data Engineer	Double-entry ledger schema, automated settlement reconciliation pipelines, data integrity controls	PostgreSQL, Redis, RabbitMQ, Docker

Code Samples

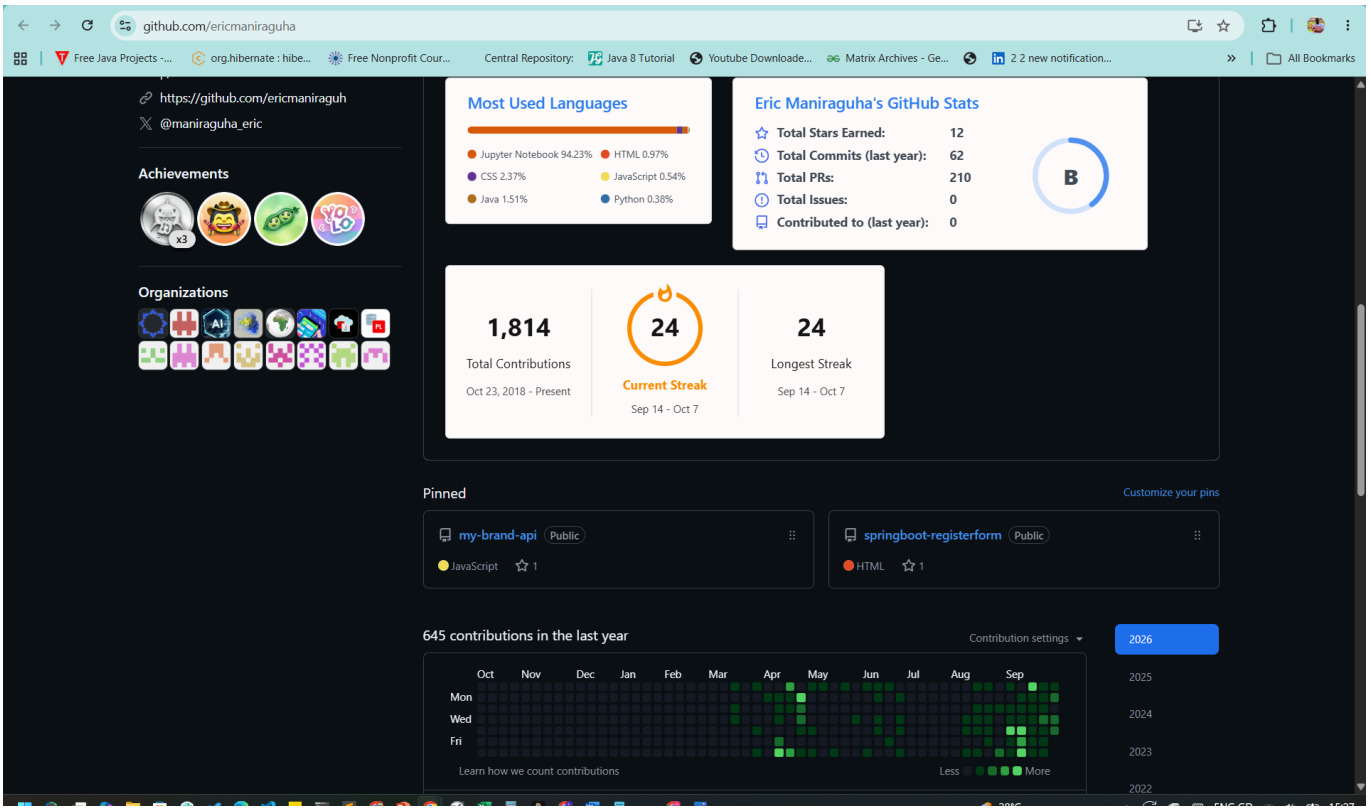
Since client code can't be shared, I wrote these samples from scratch to demonstrate the same techniques. They target **PostgreSQL 15+**.

Section	What it demonstrates
01 · Database Administration	Constraints, views, materialized views, stored procedures, audit triggers, role hierarchy, row-level security, monitoring queries
02 · Performance Tuning	Slow-query case study (identify → `EXPLAIN ANALYZE` → index → rewrite → verify), CTEs, window functions

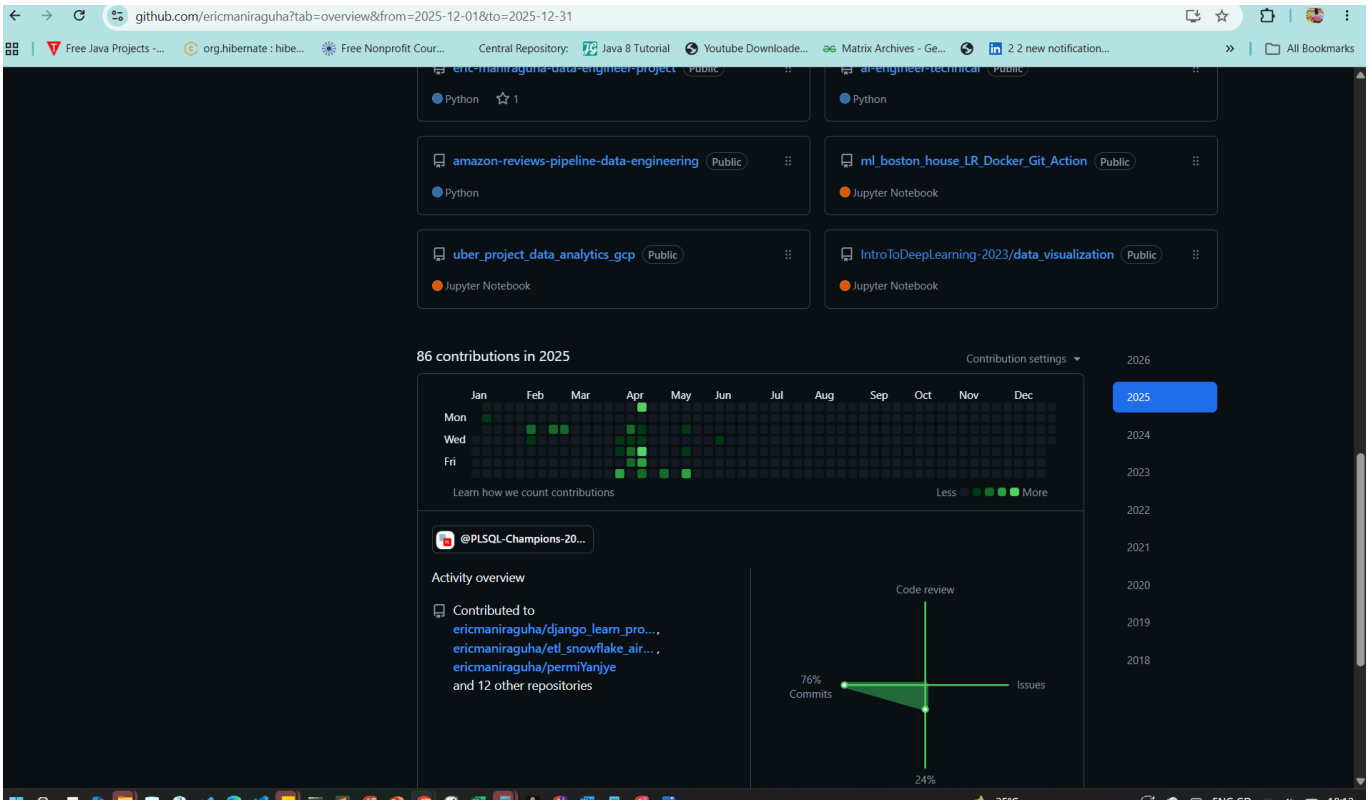
GitHub Contribution History

Most of my daily work happens in **private organizational repositories**. These graphs reflect that activity.

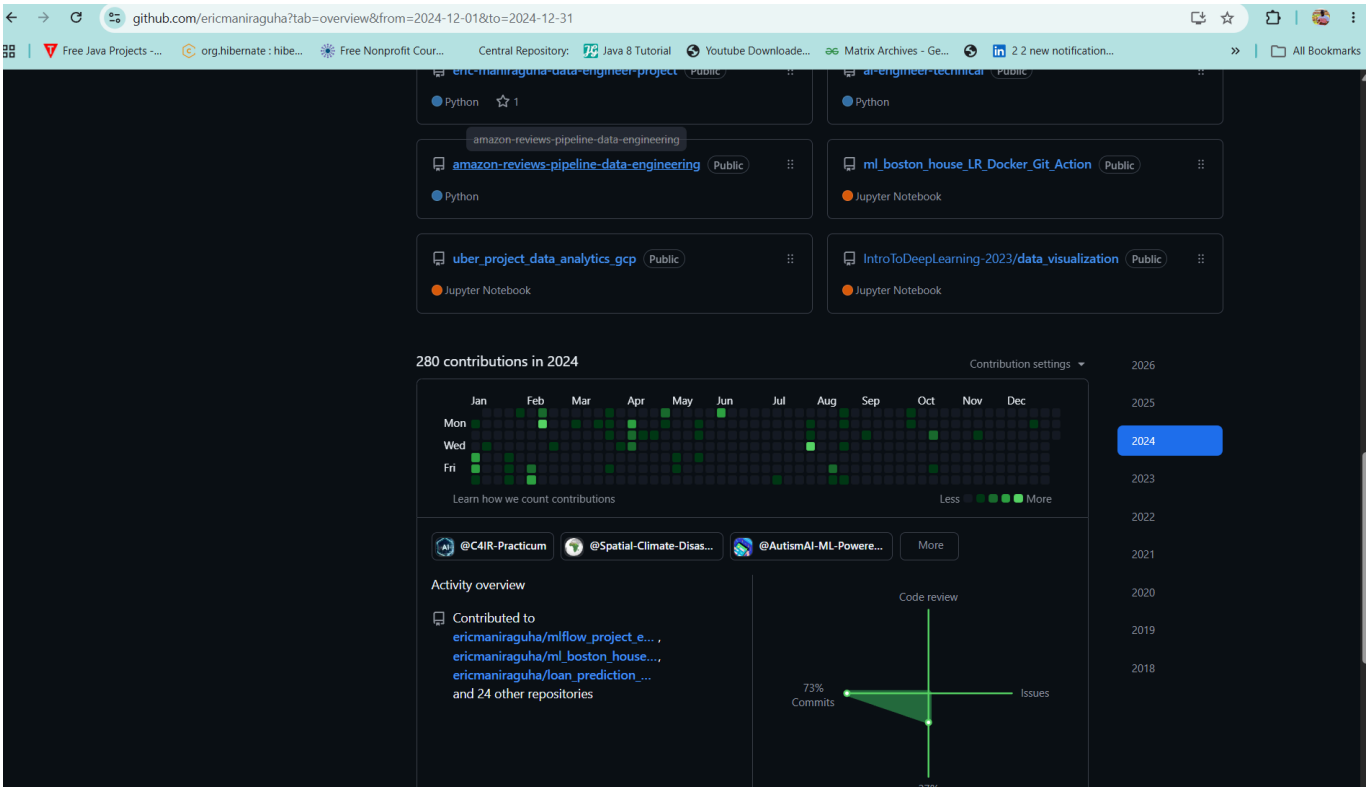
2026



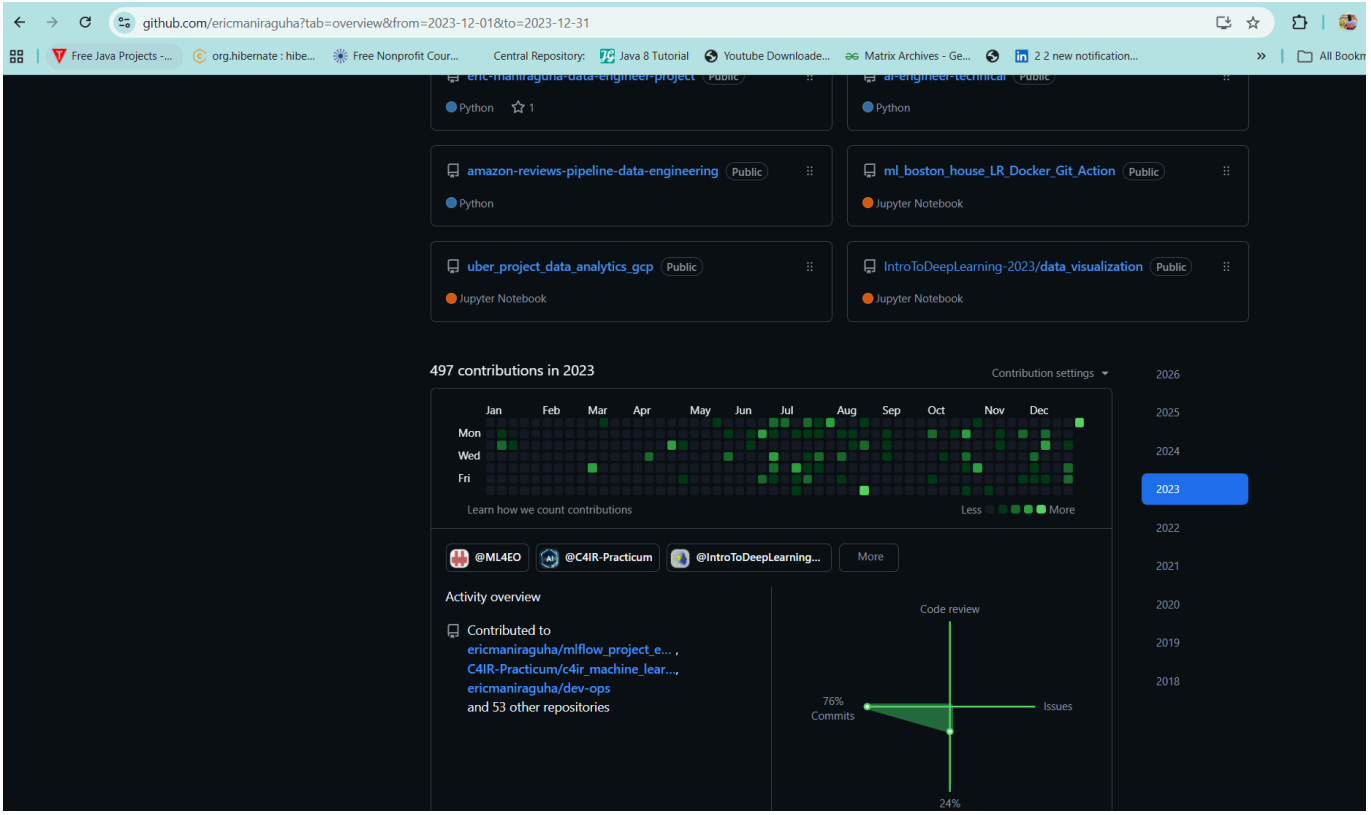
2025


github.com/ericmaniraguha?tab=overview&from=2025-12-01&to=2025-12-31
amazon-reviews-pipeline-data-engineering Public
Python
ml_boston_house_LR_Docker_Git_Action Public
Jupyter Notebook
uber_project_data_analytics_gcp Public
Jupyter Notebook
IntroToDeepLearning-2023/data_visualization Public
Jupyter Notebook
86 contributions in 2025
Contribution settings
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Mon
Wed
Fri
Learn how we count contributions
Less More
@PLSQL-Champions-20...
Activity overview
Contributed to
ericmaniraguha/django_learn_pro... ,
ericmaniraguha/etl_snowflake_air... ,
ericmaniraguha/permiYanjye
and 12 other repositories
Code review
76% Commits
Issues
24%
2026
2025
2024
2023
2022
2021
2020
2019
2018

2024


github.com/ericmaniraguha?tab=overview&from=2024-12-01&to=2024-12-31
amazon-reviews-pipeline-data-engineering Public
Python
ml_boston_house_LR_Docker_Git_Action Public
Jupyter Notebook
uber_project_data_analytics_gcp Public
Jupyter Notebook
IntroToDeepLearning-2023/data_visualization Public
Jupyter Notebook
280 contributions in 2024
Contribution settings
Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
Mon
Wed
Fri
Learn how we count contributions
Less More
@C4IR-Practicum
@Spatial-Climate-Disas...
@AutismAI-ML-Powere...
More
Activity overview
Contributed to
ericmaniraguha/mlflow_project_e... ,
ericmaniraguha/ml_boston_house...,
ericmaniraguha/loan_prediction_...
and 24 other repositories
Code review
73% Commits
Issues
2026
2025
2024
2023
2022
2021
2020
2019
2018

2023



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